

Lecture:3 Section :3 Test: 1



27. On segment WZ above, if $WY = 21$, $XZ = 26$, and YZ is twice WX , what is the value of XY ?

- (A) 5
(B) 10
(C) 11
(D) 16
(E) It cannot be determined from the information given.

30.

28. To reproduce an old photograph, a photographer charges x dollars to make a negative, $\frac{3x}{5}$ dollars for each of the first 10 prints, and $\frac{x}{5}$ dollars for each print in excess of 10 prints. If \$45 is the total charge to make a negative and 20 prints from an old photograph, what is the value of x ?

- (A) 3
(B) 3.5
(C) 4
(D) 4.5
(E) 5

solution:

$$27) xy + yz - xy - wx = 26 - 21 = 5$$

$$yz - wx = 5 \dots \dots \dots (A)$$

$$yz = 2 * wx \text{ (given)}$$

from A;

$$\text{so } 2 * wx - wx = 5$$

$$wx + xy = 21$$

$$xy = 21 - 5 = 16$$

28)

we have to find 20 prints

first 10 prints = $(3x/5)*10$
 next 10 prints = $(x/5)*10$

and negative cost = x
 $x + (3x/5)*10 + (x/5)*10 = 45$
 $x + 6x + 2x = 45$
 $9x = 45$
 $x = 5$

so the correct answer is (E)

$\triangle RST$ is isosceles and $\angle RST = 40^\circ$.

12. The sum of the measures of the two angles of $\triangle RST$ that have equal measure 120°

12) $\angle a + \angle a + 40 = 180$
 $2*\angle a = 180 - 40 = 120$
 $2*\angle a = 120$
 again
 $40 + 40 + \angle a$
 $80 + \angle a$

$120 \neq 120$ ©

$80 \neq 120$ (B)

THE CORRECT ANSWER IS D

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11. The sum of all the integers from 19 to 59, inclusive

The sum of all the integers from 22 to 60, inclusive

$19+20+21+22+\dots+59$
 $19+20+21$
 60
 (C) is the correct answer.

$22+\dots+59+60$
 60
 60

A farmer has two large plots of land that are equal in area. The first is divided into 16 parcels with n acres in each and the second is divided into 20 parcels with m acres in each.

3.

n

m

given equal area

$$\text{Area 1} = \text{Area 2}$$

$$16*n = 20*m$$

$$4*n = 5*m$$

therefore , $n > m$

(A) is the correct

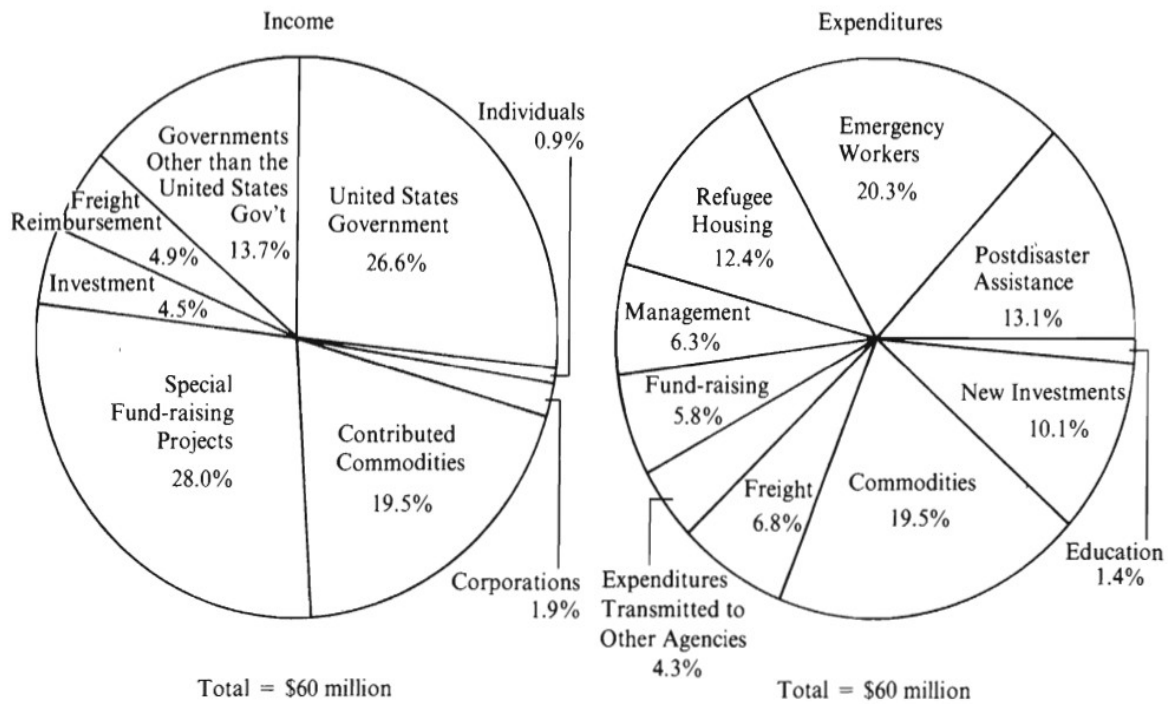
28. A widow received $\frac{1}{3}$ of her husband's estate, and each of her three sons received $\frac{1}{3}$ of the balance. If the widow and one of her sons received a total of \$60,000 from the estate, what was the amount of the estate?

- (A) \$90,000
- (B) \$96,000
- (C) \$108,000
- (D) \$135,000
- (E) \$180,000

28. let, real estate price = $3x$;
 widow received = $(\frac{1}{3}) * 3x$
 widow received = x ;
 rest = $3x - x = 2x$;
 3 sons got $2x$
 1 son received $\frac{2x}{3}$
 given, $x + \frac{2x}{3} = 60,000$ [widow and one son received = 60,000]
 $3x + 2x = 60,000 * 3$
 $x = (60,000 * 3) / 5$
 $3x = 120,000$
 $3x = 108,000$

Questions 21-25 refer to the following graphs.

INCOME AND EXPENDITURES OF AN INTERNATIONAL SERVICE AGENCY—YEAR X



21. Approximately how much of the agency's income was provided by contributed commodities ?

Total = \$60 million

contributed commodities $= (19.5/100) * 60 = 11.7 = \mathbf{12 \text{ million. (A)}}$

22. Of the following, the category that had expenditures most nearly equal to the average (arithmetic mean) expenditures per category was

Total expenditure :60 million

Total category : 10

expenditure per category : $60/10 = 6$

now % = $6/60 * 100 = 10\%$.

which is approximately expenditure of new investments (10.1%)

(D) is the correct answer.

23. Income from which of the following sources was most nearly equal to \$2.9 million?

$(2.9 \times 100) / 60 = 4.8\%$ which = Freight reimbursement 4.9%

the correct answer is (B).

24. (E)

$1/3 \times 124.6 / 100 \times 60 + 1/5 \times 20.3 / 100 \times 60 + 2/3 \times 13.1 / 100 \times 60 + 1/4 \times 19.5 \times 60 / 100$

$= (60/100) \times (\{12.4/3\} + \{20.3/5\} + \{1/4 \times 19.5\} + \{2/3 \times 13.1\})$

$= 13.07898$

correct answer is (E)

25. (paraphrase) Of the following, which is the closest approximation to the percent of freight expenditures and which is NOT covered by freight reimbursement income?

25. Freight expenditure = 6.8, Freight income = 4.9

$6.8 - 4.9 = 1.9$ not covered by income : 4.9

since it is expense so, expenditure > income.

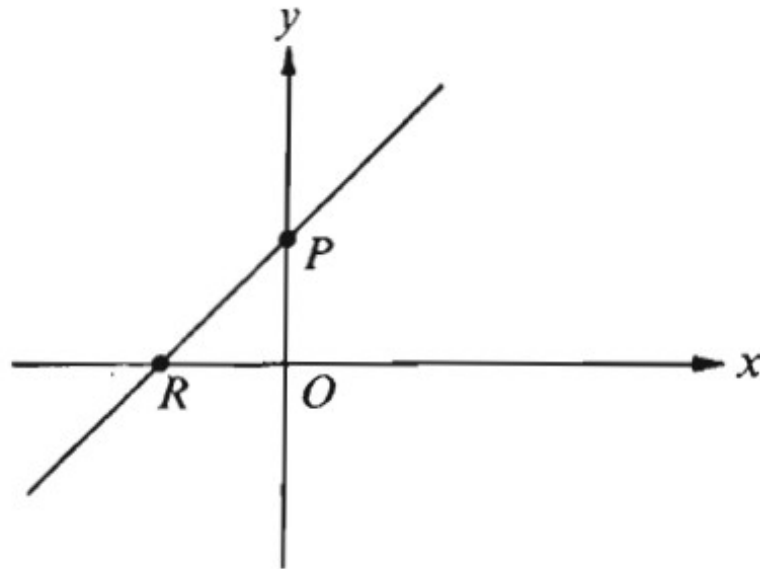
Formula % expense = $((\text{expenditure} - \text{income}) / \text{expenditure}) \times 100$

$(1.9 / 6.8) \times 100$

$= 28\%$. (B)

Column A

Column B



The equation of the line graphed on the rectangular coordinate system above is:

$$y = \frac{8x}{9} + 3$$

12.

PO

RO

$$\text{given } y = \frac{8x}{9} + 3$$

$$9y = 8x + 27$$

$$9y - 8x = 27$$

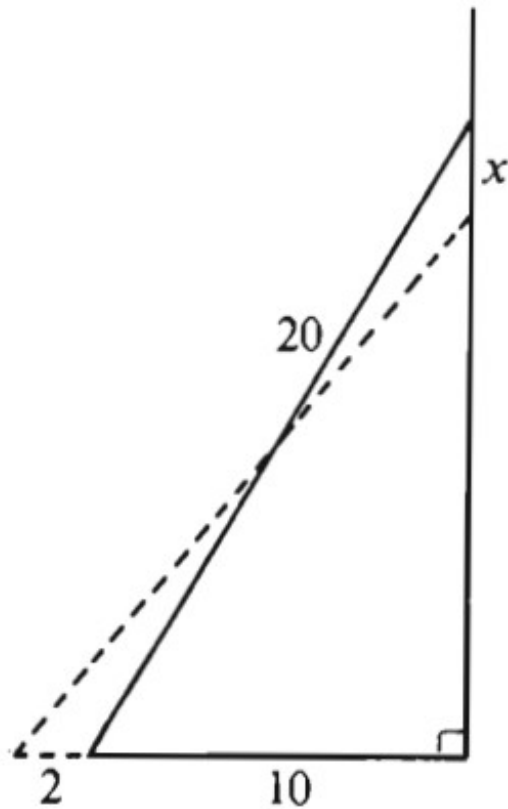
$$\frac{y}{3} - \frac{x}{\frac{27}{8}} = 1 \quad [x/a + y/b = 1]$$

now, we have to find

$$\begin{array}{ccc} \text{PO} & & \text{RO} \\ 3 & & 27/8 \\ 24 & < & 27 \text{ (B)} \end{array}$$

Column A

Column B



A 20-foot ladder leaning against a vertical wall with the base of the ladder 10 feet from the wall is pulled 2 feet farther out from the wall, causing the top of the ladder to drop x feet.

14.

x

2

$$\sqrt{400-100}$$

$$\sqrt{300}$$

$$10\sqrt{3}$$

$$10 \cdot 1.7$$

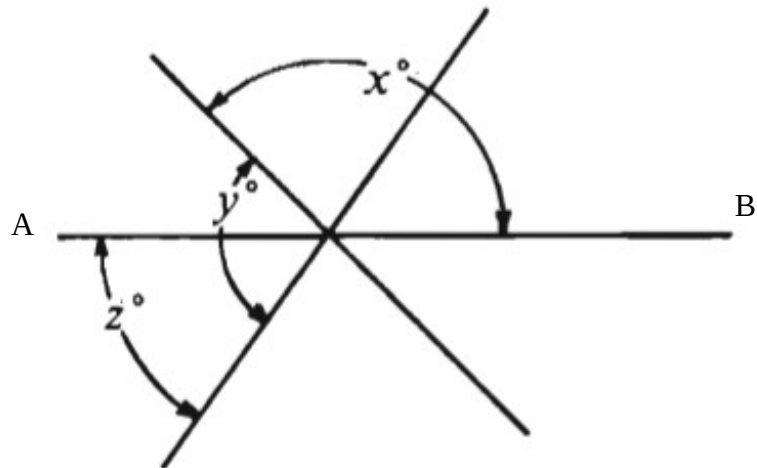
$$17$$

now,
20:12:x
5:3:x/4[3:4:5]

$x/4=4$
 $x=16$
20:12:16

according the question

17-16 <2
1<2(B)



26. In the figure above, if $x = 110$ and $y = 120$, then $z =$

- (A) 10
- (B) 40
- (C) 50
- (D) 60
- (E) 70

$$X+Y=110+120=230$$

Angle between A and B is 180

$$z=230-180=50(\text{C})$$